

E-Learning System.

Build a **basic E-Learning System** using Laravel that helps you understand how real applications handle:

- Authentication
- Role Management
- Authorization
- CRUD operations
- File uploads

Do not focus on the UI/UX. I would like you to focus on functionality and Laravel best practices.

Project Overview

Create a small **E-Learning Platform** with:

- **Admins**
- **Instructors**
- **Students**

Users can log in, instructors can create courses, students can enroll, and view content.

Authentication

Requirements

- Use Laravel authentication (Breeze, Jetstream, or Laravel UI)
- Users must be able to:
 - Register
 - Login
 - Logout

User Fields

- Name
- Email
- Password

- Role (admin, instructor, student)

Role Management

Roles & Permissions

Admin

- Can view all users
- Can change a user's role
- Can delete users

Instructor

- Can create, edit, delete **their own courses**
- Can upload course materials

Student

- Can view available courses
- Can enroll in a course
- Can view course materials (read-only)

Authorization

Use **Laravel Policies or Gates** to enforce rules.

Examples:

- Only instructors can create courses
- Instructors can only edit **their own** courses
- Students cannot access course creation pages
- Only enrolled students can view course content

CRUD Features

Courses (Main CRUD)

Each course should have:

- Title

- Description
- Instructor (user_id)
- Cover image (file upload)

Operations:

- Create course
- View course
- Edit course
- Delete course

Enrollments

- Students can enroll in a course
- Prevent duplicate enrollments
- Store relationship in a pivot table

File Uploads

Course Materials

- Instructors upload:
 - PDFs
 - Videos (optional)
- Files stored using Laravel Storage (storage/app/public)
- Each file linked to a course

Students:

- Can view/download materials
- Cannot upload or delete files

Database Design (Expected Tables)

Minimum tables:

- users
- courses
- enrollments
- materials (course files)

Use:

- Migrations
- Eloquent relationships

Laravel Concepts You Must Use

- Routes (web.php)
- Controllers
- Form Requests (validation)
- Middleware (role checks)
- Policies or Gates
- Blade templates
- Eloquent relationships
- Storage facade for uploads

Expectations (Very Important)

By the end of this task, you should be able to:

1. **Explain** how authentication works in Laravel
2. **Demonstrate** role-based access control
3. Build full **CRUD** without scaffolding shortcuts
4. Use **authorization properly** (not just hiding buttons)

5. Upload and retrieve files securely
6. Understand relationships (hasMany, belongsTo, belongsToMany)
7. Push the project to GitHub with:
 - a. Clear commits
 - b. README explaining the system